CENTRAL BOARD OF SECONDARY EDUCATION

SAMPLE QUESTION PAPER

Academic Session: 2026–27

Class: 12

Subject: Physics

Maximum Marks: 50

Time Allowed: 2 hours

GENERAL INSTRUCTIONS

• There are 5 sections in the question paper. All questions are compulsory.

• Section A contains 10 Multiple Choice Questions (MCQs) of 1 mark each.

• Section B contains 4 Very Short Answer (VSA) questions of 2 marks each.

• Section C contains 3 Short Answer (SA) questions of 3 marks each.

• Section D contains 1 Long Answer (LA) question of 5 marks.

• Section E contains 2 Case Study questions of 4 marks each.

• There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C and one question in Section D. You have to attempt only one of the choices in such questions.

• Use of calculators is not permitted.

SECTION A

Multiple Choice Questions

Q1. An electric dipole is placed in a uniform electric field. The net force on the dipole is:

(A) Always zero

(B) Never zero

(C) Depends on the orientation of the dipole

(D) Depends on the magnitude of the electric field

(1 Mark)

Q2. The magnetic field lines inside a current-carrying solenoid are:

(A) Circular and concentric

(B) Straight and parallel to the axis

(C) Straight and perpendicular to the axis

(D) Helical

(1 Mark)

Q3. Which of the following phenomena supports the wave nature of light?

(A) Photoelectric effect

(B) Compton effect

(C) Interference

(D) Blackbody radiation

(1 Mark)

Q4. The energy of a photon of wavelength λ is:

(A) $hc\lambda$

(B) $h\lambda/c$

(C) hc/λ

(D) λ/hc

(1 Mark)

Q5. A p-n junction diode acts as a:

(A) Conductor

(B) Insulator

(C) Unidirectional switch

(D) Bidirectional switch

(1 Mark)

Q6. The half-life of a radioactive substance is 10 days. After 20 days, the fraction of the original sample remaining will be:

(A) 1/2

(B) 1/4

(C) 1/8

(D) 1/16

(1 Mark)

Q7. The SI unit of magnetic flux is:

(A) Tesla

(B) Weber

(C) Henry

(D) Ampere-meter

(1 Mark)

Q8. When a charged particle enters a uniform magnetic field at an angle, its path will be:

(A) Straight line

(B) Circle

(C) Helix

(D) Parabola

(1 Mark)

Q9. The phenomenon of total internal reflection occurs when light travels from:

(A) Denser to rarer medium

(B) Rarer to denser medium

(C) Any medium to any other medium

(D) Vacuum to any medium

(1 Mark)

Q10. In an AC circuit containing only an inductor, the phase difference between voltage and current is:

(A) 0°

(B) 45°

(C) 90°

(D) 180°

(1 Mark)

SECTION B

Very Short Answer Questions

Q11. Define electric potential. Is it a scalar or a vector quantity?

(2 Marks)

Q12. State Faraday's laws of electromagnetic induction.

(2 Marks)

Q13. What is the de Broglie wavelength of an electron moving with a speed of 1.0×10^6 m/s? (Given: mass of electron = 9.1×10^{-31} kg, Planck's constant $h = 6.63 \times 10^{-34}$ Js)

(2 Marks)

Q14. Distinguish between intrinsic and extrinsic semiconductors.

OR

Draw the circuit diagram of a half-wave rectifier and explain its working.

(2 Marks)

SECTION C

Short Answer Questions

Q15. Derive an expression for the capacitance of a parallel plate capacitor with a dielectric slab of thickness t ($t < d$) inserted between the plates.

(3 Marks)

Q16. Explain the working of a moving coil galvanometer. Why is its scale linear?

(3 Marks)

Q17. Using Huygens' principle, explain the phenomenon of refraction of light.

OR

Derive the lens maker's formula for a thin lens.

(3 Marks)

SECTION D

Long Answer Questions

Q18. State Bohr's postulates for the hydrogen atom. Derive an expression for the radius of the n-th orbit of the electron in a hydrogen atom.

OR

Explain the formation of energy bands in solids. On the basis of energy bands, distinguish between conductors, insulators, and semiconductors.

(5 Marks)

SECTION E

Case Study

Q19. Read the following paragraph and answer the questions that follow:

A series LCR circuit is connected to an AC source. The circuit consists of an inductor (L), a capacitor (C), and a resistor (R) connected in series. The impedance of the circuit depends on the frequency of the AC source. At a particular frequency, known as the resonant frequency, the inductive reactance becomes equal to the capacitive reactance, leading to a minimum impedance and maximum current in the circuit. This phenomenon is called resonance.

(a) What is the condition for resonance in a series LCR circuit?

(b) Write the expression for the resonant frequency (f■) of a series LCR circuit.

(c) If L = 100 mH, C = 10 μF, and R = 50 Ω, calculate the resonant frequency of the circuit.

(d) What is the phase difference between voltage and current at resonance?

(4 Marks)

Q20. Read the following paragraph and answer the questions that follow:

The photoelectric effect is the emission of electrons when electromagnetic radiation, such as light, hits a material. Electrons emitted in this manner are called photoelectrons. The energy of the emitted photoelectrons depends on the frequency of the incident light, not its intensity. There is a minimum frequency, called the threshold frequency, below which no photoelectrons are emitted, regardless of the intensity of the light. This phenomenon was explained by Albert Einstein using Planck's quantum theory.

(a) What is the photoelectric effect?

(b) What is meant by threshold frequency?

(c) If the work function of a metal is 3.0 eV, what is the maximum wavelength of light that can cause photoelectron emission from this metal? (Given: h = 6.63 × 10^-34 Js, c = 3 × 10^8 m/s, 1 eV = 1.6 × 10^-19 J)

(d) How does the kinetic energy of photoelectrons change with the intensity of incident light?

(4 Marks)

End of Question Paper